SCOPE MANAGEMENT PLAN

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

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INTRODUCTION

The Scope Management Plan provides the scope framework for this project. This plan documents the scope management approach; roles and responsibilities as they pertain to project scope; scope definition; verification and control measures; scope change control; and the project's work breakdown structure. Any project communication which pertains to the project's scope should adhere to the Scope Management Plan.

This project involves the design, development, and testing of LitPath AI: Smart Pathfinder for Theses and Dissertations, an AI-powered search engine built to enhance access to the DOST-STII Library's collection of graduate theses and dissertations. The system is developed as a web-based application hosted on the HecTech team's own server infrastructure, integrating with the existing DOST-STII Online Public Access Catalog (OPAC) without replacing it. This includes the design of the user interface, all backend and frontend programming, AI integration, and system testing and validation. Development is carried out entirely by the four-member HecTech team, with no external outsourcing anticipated.

SCOPE MANAGEMENT APPROACH

For the LitPath AI project, scope management is the primary responsibility of the Project Manager, Tracie Tomon. The scope for this project is defined by the Scope Statement, the Work Breakdown Structure (WBS), and the WBS Dictionary, all of which are grounded in the requirements collected through stakeholder interviews with DOST-STII, the Design Thinking process, and the Project Charter authorized by the Project Sponsor, Khasian Eunice Romulo.

The Project Manager, Sponsor, and stakeholders will establish and approve documentation for measuring project scope, which includes deliverable quality checklists and work performance measurements tracked through OpenProject.

Proposed scope changes may be initiated by the Project Manager, any member of the HecTech development team, or the Project Sponsor. All change requests must be submitted to the Project Manager, who will evaluate the request and conduct an impact assessment. Upon initial acceptance by the Project Manager, the scope change will be submitted to the Project Sponsor for final approval. Once approved, the Project Manager will update all project documents and communicate the scope change to all team members and stakeholders.

The Project Sponsor, Khasian Eunice Romulo, is responsible for the formal acceptance of all final project deliverables and the overall project scope.

ROLES AND RESPONSIBILITIES

The Project Manager, Sponsor and team will all play key roles in managing the scope of this project. As such, the project sponsor, manager, and team members must be aware of their responsibilities in order to ensure that work performed on the project is within the established

scope throughout the entire duration of the project. The table below defines the roles and responsibilities for the scope management of this project.

Name	Role	Responsibilities
Khasian Eunice Romulo	Client / Project Sponsor	- Approve or deny scope change requests as appropriate - Evaluate need for scope change requests - Accept project deliverables
Tracie Tomon	Project Manager/ Team lead	- Measure and verify project scope - - Facilitate and validate scope change requests - Conduct and participate in impact assessments - Organize and lead change control meetings - Facilitate team-level change review process - Communicate outcomes of scope change requests to stakeholders and team - Update project documents upon approval of scope changes -
Jenine Elaine Dulay	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary
Charijoy Cempron	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary
Joshua Sta Ana	Team Member	- Participate in defining change resolutions - Evaluate the need for scope changes and communicate them to the project manager as necessary

Table 1.1, *Scope Management Roles and Responsibilities*

SCOPE DEFINITION

The scope for the LitPath AI project was defined through a comprehensive requirements collection process rooted in both the client's needs and the existing system's limitations. First, a thorough analysis was performed on the DOST-STII Library's current Online Public Access Catalog (OPAC) system based on feedback gathered through stakeholder interviews with DOST-STII

librarians, staff, student researchers, and DOST employees during the Design Thinking Empathize stage. From this information, the project team identified the core problems: exact keyword matching, limited filtering options, slow search performance, and the lack of AI-powered features and citation tools.

Building on this, the project requirements were documented, and the scope was formally established through the Project Charter (authorized March 29, 2026) and the Scope Statement included in the final project paper. The project description and deliverables were further refined based on input from the client (DOST-STII), the PBL instructor, and the team's own expert judgment in software design, AI integration, and user interface development.

The tools and techniques used to define the project scope include stakeholder interviews, the 5 WHYs analysis, persona development, How-Might-We exercises, Crazy 8s ideation, Desirability-Viability-Feasibility analysis, and prototype testing — all carried out across the five stages of the Design Thinking process.

PROJECT SCOPE STATEMENT

Product Scope Description

LitPath AI is an AI-powered search engine designed to improve access to the DOST-STII Library's collection of graduate theses and dissertations. It enhances the existing OPAC system by offering AI-powered query analysis, an automated citation generator, bookmarking, research history, follow-up questions, and an admin dashboard for library staff to monitor usage analytics.

Product Acceptance Criteria

The project will be accepted as complete when the following criteria are met:

1. The system achieves a minimum of 70% relevance accuracy for its top 10 search results, validated through user feedback and expert review.
2. The system achieves an average user satisfaction score of at least 3.5 out of 5 from usability testing participants (student researchers and DOST employees).
3. The system consistently processes search queries and displays results in under one second.
4. The system is fully integrated with the existing DOST-STII OPAC.
5. All key deliverables have been formally reviewed and accepted by the Project Sponsor.

Project Deliverables

1. AI-powered search engine — a fully functional backend system providing semantic search and content-based recommendations, delivered with an API for integration into the existing OPAC.
2. User Interface — a responsive web interface, a built-in automated citation generator supporting APA, MLA, Chicago, and IEEE styles and bookmark feature.

3. User Feedback Tool — a mechanism for capturing star ratings and thumbs up/down and comments from users to measure satisfaction and validate search relevance.
4. Admin Dashboard — an interface for DOST-STII library staff to track search trends and monitor system usage.
5. Manual and Final Report — complete technical documentation, a user guide, and a final report verifying that all project objectives have been met.

Project Constraints

1. The project must be deployed by June–July 2026 in alignment with the academic timeline.
2. The system will be hosted on a local server maintained by the HecTech team, eliminating external hosting costs. Any AI/LLM API expenses during production will be coordinated with DOST-STII, as no development budget has been allocated to the team.
3. Development is limited to the four-member HecTech team, and no external outsourcing is anticipated.

WORK BREAKDOWN STRUCTURE

To effectively manage the work required to complete this project, it will be subdivided into individual work packages which will not exceed 40 hours of work. This will allow the Project Manager to more effectively manage the project's scope as the project team works on the tasks necessary for project completion. The project is broken down into three phases: the design phase; the programming phase; and the testing phase. Each of these phases is then subdivided further down to work packages which will require no more than 40 hours of work and no less than 4 hours of work (see WBS structure below).

To effectively manage the work required to complete LitPath AI, the project has been subdivided into individual work packages assigned to each team member and tracked in OpenProject. Work packages are organized by project phase and are scoped to be manageable within each sprint cycle.

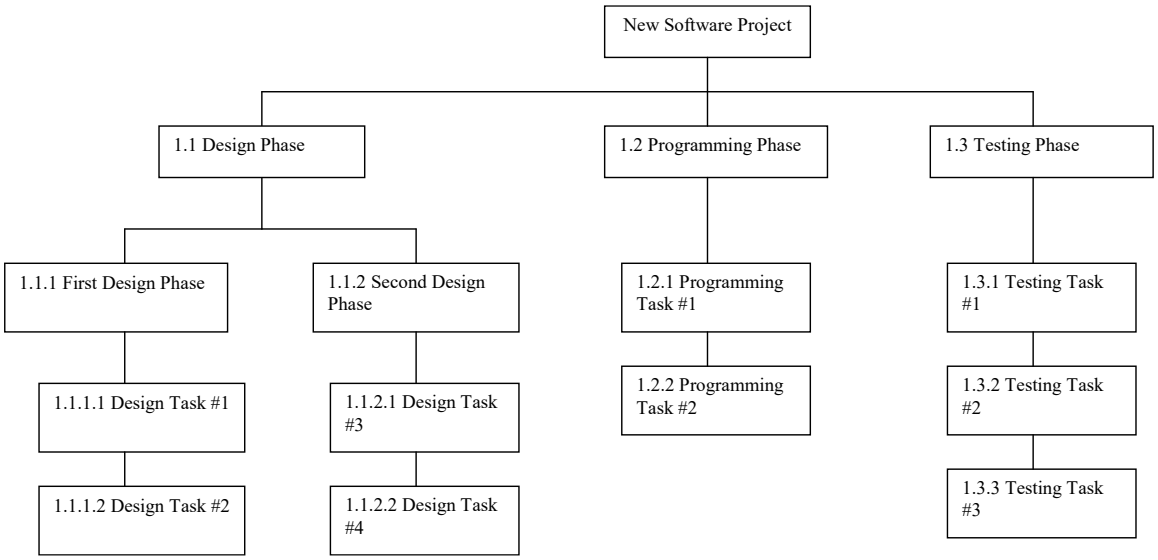

Figure 1.1, *Work Breakdown Structure (WBS)*

In order to more clearly define the work necessary for project completion the WBS Dictionary is used. The WBS Dictionary includes an entry for each WBS element. The WBS Dictionary includes a detailed description of work for each element and the deliverables, budget and resource needs for that element. The project team will use the WBS Dictionary as a statement of work for each WBS element.

AY 2025-2026 T3 Sprint 01 | Feedback Management System

Sprint Period: 03/23/2026 – 04/03/2026 **Total Points:** 80 **Total Features:** 12

Level	WBS Code	Element Name	Definition
1	**1**	**Feedback Management System**	All work to implement and deliver features for the Feedback Management System under AY 2025-2026 T3 Sprint 01.
2	**1.1**	**AY 2025-2026 T3 Sprint 01**	Sprint covering the period 03/23/2026 – 04/03/2026 with a total of 80 story points across 12 feature items.
3	1.1.1	Feature 1.0 User Account Registration System	Implement the user account registration system including sign-up forms, input validation, account creation logic, and confirmation workflows. (8 points, Issue #4674)
3	1.1.2	Feature 2.0 LLM Source Accuracy Optimization / Enhance Prompt	Optimize the accuracy of LLM-generated responses by refining source selection logic and enhancing prompt

Level	WBS Code	Element Name	Definition
			engineering techniques to improve output quality and relevance. (10 points, Issue #4679)
3	1.1.3	Feature 3.0 Welcome Page & Form UI Modernization	Redesign and modernize the welcome page and associate form interfaces to improve user experience, visual consistency, and accessibility standards. (5 points, Issue #4689)
3	1.1.4	Feature 4.0 Display Philippine Standard Time (PST)	Update all date and time displays throughout the application to reflect Philippine Standard Time (PST, UTC+8), ensuring consistent localized timestamps across the system. (5 points, Issue #4690)
3	1.1.5	Feature 5.0 Clickable Modal Interface (Feedback Manager)	Implement a clickable modal interface within the Feedback Manager module to allow users to view detailed feedback entries in an overlay dialog without leaving the current page context. (5 points, Issue #4692)
3	1.1.6	Feature 6.0 UI Decluttering (Feedback Table View)	Streamline and declutter the Feedback Table View by removing redundant elements, reorganizing columns, and improving the overall layout for enhanced readability and usability. (7 points, Issue #4693)
3	1.1.7	Feature 7.0 Data Pagination (Feedback Manager)	Introduce data pagination functionality to the Feedback Manager to efficiently handle large datasets, enabling users to navigate through records in manageable pages. (7 points, Issue #4694)
3	1.1.8	Feature 8.0 Targeted Dormant Material Export	Develop an export feature that allows administrators to selectively export dormant materials from the system based on defined criteria, supporting data management and archival processes. (8 points, Issue #4695)
3	1.1.9	Feature 9.0 Failed Query Keyword Tracking	Implement a tracking mechanism to capture and log keywords from failed search queries, enabling the team to identify content gaps and improve system coverage over time. (8 points, Issue #4696)
3	1.1.10	Feature 10.0 "Least Viewed" Metric Tracking	Build a metric tracking system that identifies and monitors the least-viewed content items, providing insights to inform content curation and resource prioritization decisions. (8 points, Issue #4697)
3	1.1.11	Feature 11.0 Column Merger & Analysis Dropdown Refinement	Merge related data columns and refine the analysis dropdown menus to reduce redundancy, improve navigation clarity, and streamline the analytical workflow for end users. (4 points, Issue #5100)

Level	WBS Code	Element Name	Definition
3	1.1.12	Feature 12.0 Explicit Date Formatting & "Feedback Type" Column Renaming	Standardize date formatting across the application with explicit format definitions and rename the Feedback Type column for improved clarity and consistency in the data presentation layer. (5 points, Issue #5101)

All features have a status of Closed as of the sprint end date.

Scope Verification

As the LitPath AI project progresses, the Project Manager will verify interim project deliverables against the original scope as defined in the Scope Statement, WBS, and WBS Dictionary. Once the Project Manager confirms that a deliverable meets the requirements defined in the project plan, the Project Manager and Project Sponsor will meet for formal acceptance.

During these milestone review meetings, the Project Manager will present the deliverable to the Project Sponsor, Khasian Eunice Romulo, for formal acceptance. The Project Sponsor will accept the deliverable by signing a project deliverable acceptance document. Key verification checkpoints are aligned with the milestone schedule:

- Completion of the UI and core features (Oct–Nov 2025) ✓
- Completion of AI Search Engine and advanced features (Oct 2025–Jan 2026) ✓
- Feature refinement and client feedback integration (Feb–Mar 2026) ✓
- Internal testing and temporary deployment (Apr 2026) — In-progress
- Usability testing results (May–Jun 2026) — Not started
- Final deployment and handover (Jun–Jul 2026) — Not started

This incremental verification approach ensures that project work remains within scope on a consistent basis throughout the project lifecycle rather than being evaluated only at the end.

Scope Control

The Project Manager and the HecTech development team will work together to control the scope of the LitPath AI project. The team will use the WBS Dictionary as the statement of work for each WBS element, ensuring that only the described work is performed and that defined deliverables are produced for each element. The Project Manager will oversee the team and the progression of the project to ensure this scope control process is followed throughout all phases.

If a change to the project scope is needed, the following process must be carried out:

1. Any team member, the Project Manager, or the Project Sponsor may identify and initiate a scope change request.

2. The change request must be submitted to the Project Manager in writing (via Viber or a formal change request document stored on SharePoint).
3. The Project Manager will review the request and conduct an impact assessment, evaluating effects on schedule, resources, and the June–July 2026 deployment deadline.
4. If the Project Manager denies the request as outside the intent of the project, the requestor will be notified with a rationale.
5. If the Project Manager approves the request at the initial level, a change control meeting will be convened with the team and Project Sponsor to review the change further.
6. Upon approval by the Project Sponsor, Khasian Eunice Romulo, the Project Manager will update all project documents (Scope Statement, WBS, WBS Dictionary, OpenProject tasks) and communicate the approved scope change to all team members and stakeholders.

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

_____ Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)